KOSZUL-REGULAR, H1-REGULAR AND QUASI-REGULAR IMMERSIONS ARE FPQC LOCAL ON THE BASE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof displays the reformulation of the descent hypothesis as a free-standing algebra statement; that reformulation is the load-bearing part of the span and is reproduced verbatim and recorded as an obligation.

Editorial note. The counted spans carry 4 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The criterion for a property of morphisms to be fpqc local on the base. Same chapter, Lemma lemma-descending-properties-morphisms, invoked at proof lines 6499-6501. Reduces the assertion to Zariski locality, stability under flat base change and the descent statement to be translated.
- Koszul-regular, H1-regular and quasi-regular immersions and their Zariski locality. Divisors, Definition divisors-definition-regular-immersion, invoked at proof lines 6501-6504. Supplies the definition and the Zariski locality that the criterion asks for.
- Flat base change preserves Koszul-regular, H1-regular and quasi-regular immersions. Divisors, Lemma divisors-lemma-flat-base-change-regular-immersion, invoked at proof lines 6504-6507. Gives the second hypothesis of the criterion.
- Descent of the property of an ideal to be locally generated by a regular sequence along a faithfully flat ring map. More on Algebra, Lemma more-algebra-lemma-flat-descent-regular-ideal, invoked at proof lines 6515-6516. Discharges the third hypothesis once it has been reformulated as an algebra statement.

Source reference. Original definition 063J: sources/stacks-chow/tags.txt, source lines 7833–7833. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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Source reference. Original lemma 067P: sources/stacks-chow/tags.txt, source lines 7978–7978. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02KP: sources/stacks-descent/pinned-original-source.tex, source lines 5624–5624. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 068N: sources/stacks-chow/tags.txt, source lines 8012–8012. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *The properties*

$\mathcal{P}(f) =$ "*f is a Koszul-regular immersion*",
$\mathcal{P}(f) =$ "*f is an H_1-regular immersion*", *and*
$\mathcal{P}(f) =$ "*f is a quasi-regular immersion*"

are fpqc local on the base.

Proof. We will use the criterion of Lemma 02KP to prove this. By Divisors, Definition 063J being a Koszul-regular (resp. H_1-regular, quasi-regular) immersion is Zariski local on the base. By Divisors, Lemma 067P being a Koszul-regular (resp. H_1-regular, quasi-regular) immersion is preserved under flat base change. The final hypothesis (3) of Lemma 02KP translates into the following algebra statement: Let $A \to B$ be a faithfully flat ring map. Let $I \subset A$ be an ideal. If IB is locally on $\mathrm{Spec}(B)$ generated by a Koszul-regular (resp. H_1-regular, quasi-regular) sequence in B, then $I \subset A$ is locally on $\mathrm{Spec}(A)$ generated by a Koszul-regular (resp. H_1-regular, quasi-regular) sequence in A. This is More on Algebra, Lemma 068N. □